SEMM Primer - Dynamics

Sebin Oh
(sebin.oh@berkeley.edu)

University of California, Berkeley

August 20, 2026

Goals of the session

By the end of this session, you should be able to:

- Explain how (static) structural analysis is extended to structural dynamics.
- Interpret the roles of $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ in the equation of motion.
- Explain the concepts of natural frequencies, mode shapes, and modal superposition.
- Identify the mechanics, ODE, linear-algebra, and structural-analysis ideas to review before CE 225.

Goals of the session

By the end of this session, you should be able to:

- Explain how (static) structural analysis is extended to structural dynamics.
- Interpret the roles of $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ in the equation of motion.
- Explain the concepts of natural frequencies, mode shapes, and modal superposition.
- Identify the mechanics, ODE, linear-algebra, and structural-analysis ideas to review before CE 225.

Goals of the session

By the end of this session, you should be able to:

- Explain how (static) structural analysis is extended to structural dynamics.
- Interpret the roles of $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ in the equation of motion.
- Explain the concepts of natural frequencies, mode shapes, and modal superposition.
- Identify the mechanics, ODE, linear-algebra, and structural-analysis ideas to review before CE 225.

Goals of the session

By the end of this session, you should be able to:

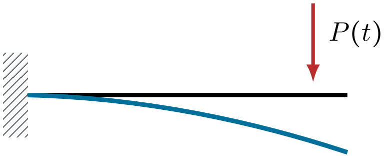
- Explain how (static) structural analysis is extended to structural dynamics.
- Interpret the roles of $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ in the equation of motion.
- Explain the concepts of natural frequencies, mode shapes, and modal superposition.
- Identify the mechanics, ODE, linear-algebra, and structural-analysis ideas to review before CE 225.

Contents

Contents

When does a structural problem become dynamic?

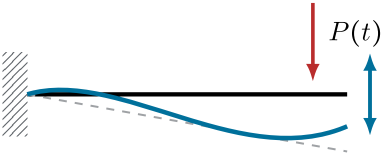
Apply the load slowly



the structure follows the load

Static / quasi-static

Change the load rapidly

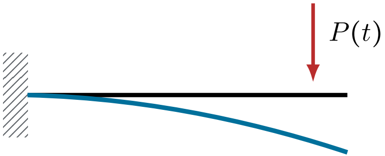


the structure keeps moving

Dynamic

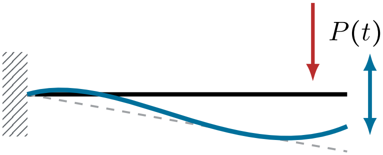
When does a structural problem become dynamic?

Apply the load slowly



the structure follows the load
Static / quasi-static

Change the load rapidly



the structure keeps moving
Dynamic

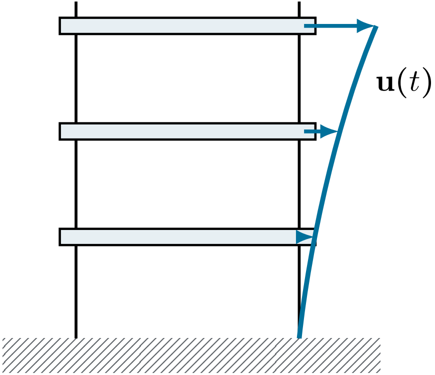
How fast is "fast"?
It depends on how quickly the structure itself tends to respond. We will come back to this.

Dynamics ... for what?

The design objectives are familiar:

- displacements and story drifts,
- deformations and strains,
- internal forces and stresses,
- accelerations of floors and equipment.

These are the same types of quantities sought in static structural analysis, but now they vary with time.

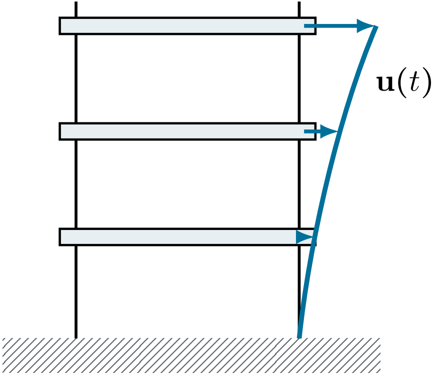


Dynamics ... for what?

The design objectives are familiar:

- displacements and story drifts,
- deformations and strains,
- internal forces and stresses,
- accelerations of floors and equipment.

These are the same types of quantities sought in static structural analysis, but now they vary with time.



Structural dynamics asks how motion and internal actions evolve under time-varying loading.

Static analysis is embedded in the dynamic equation

Static equilibrium

$$\mathbf{K}\mathbf{u} = \mathbf{p}$$

- Assemble the stiffness matrix $\mathbf{K}$ (or $\mathbf{F} = \mathbf{K}^{-1}$) from the structural model.
- Use it to map between **forces and displacements**.

$\Longrightarrow$

add inertia, damping, and time dependence

Dynamic equilibrium

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{p}(t)$$

- The same stiffness model $\mathbf{K}$ is retained.
- Inertia and damping are added to describe the response evolving in time.

$\mathbf{K}\mathbf{u}$	$\mathbf{M}\ddot{\mathbf{u}}$	$\mathbf{C}\dot{\mathbf{u}}$	$\mathbf{p}(t)$
restoring force	inertia force	damping force	external excitation

Static analysis is embedded in the dynamic equation

Static equilibrium

$$\mathbf{K}\mathbf{u} = \mathbf{p}$$

- Assemble the stiffness matrix $\mathbf{K}$ (or $\mathbf{F} = \mathbf{K}^{-1}$) from the structural model.
- Use it to map between **forces and displacements**.

$$\Longrightarrow$$

add inertia,
damping,
and time
dependence

Dynamic equilibrium

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{p}(t)$$

- The same stiffness model $\mathbf{K}$ is retained.
- Inertia and damping are added to describe the response evolving in time.

$\mathbf{K}\mathbf{u}$	$\mathbf{M}\ddot{\mathbf{u}}$	$\mathbf{C}\dot{\mathbf{u}}$	$\mathbf{p}(t)$
restoring force	inertia force	damping force	external excitation

The defining feature of dynamics is inertia

Can the inertia term be neglected?

$\mathbf{M}\ddot{\mathbf{u}}$

Yes ↓

Inertia is negligible

$\mathbf{K}\mathbf{u} \approx \mathbf{p}(t)$

Static / quasi-static

↓ No

Inertia must be retained

$\mathbf{M}\ddot{\mathbf{u}}$ affects the response

Dynamic

What about damping? The term $\mathbf{C}\dot{\mathbf{u}}$ dissipates energy and modifies the amplitude and decay of motion, but it is **not** what makes the problem dynamic. An undamped system $(\mathbf{C} = \mathbf{0})$ can still be dynamic.

The defining feature of dynamics is inertia

Can the inertia term be neglected?

$\mathbf{M}\ddot{\mathbf{u}}$

Yes ↓ | ↓ No

Inertia is negligible

$\mathbf{K}\mathbf{u} \approx \mathbf{p}(t)$

Static / quasi-static

Inertia must be retained

$\mathbf{M}\ddot{\mathbf{u}}$ affects the response

Dynamic

What about damping? The term $\mathbf{C}\dot{\mathbf{u}}$ dissipates energy and modifies the amplitude and decay of motion, but it is **not** what makes the problem dynamic. An undamped system $(\mathbf{C} = \mathbf{0})$ can still be dynamic.

The defining feature of dynamics is inertia

Can the inertia term be neglected?

$\mathbf{M}\ddot{\mathbf{u}}$

Yes ↓ | ↓ No

Inertia is negligible

$\mathbf{K}\mathbf{u} \approx \mathbf{p}(t)$

Static / quasi-static

Inertia must be retained

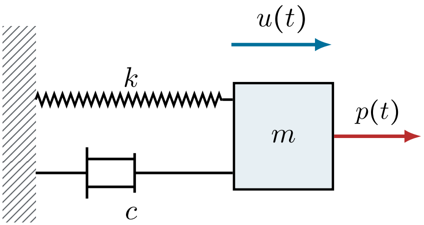
$\mathbf{M}\ddot{\mathbf{u}}$ affects the response

Dynamic

What about damping? The term $\mathbf{C}\dot{\mathbf{u}}$ dissipates energy and modifies the amplitude and decay of motion, but it is **not** what makes the problem dynamic. An undamped system $(\mathbf{C} = \mathbf{0})$ can still be dynamic.

Contents

The simplest structural oscillator



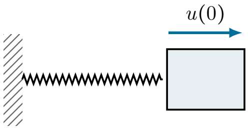
$$m\ddot{u} + c\dot{u} + ku = p(t)$$

m	inertia
c	damping / energy dissipation
k	stiffness / restoring force
$u(t)$	displacement coordinate

The single-degree-of-freedom (SDOF) oscillator is simple enough to solve analytically, yet it contains the central ideas of linear structural dynamics.

Free vibration reveals natural structural properties

Displace, release, and observe



then remove the force

$$m\ddot{u} + ku = 0$$

$$\underbrace{\omega_n = \sqrt{\frac{k}{m}}}_{\text{natural circular frequency}} \quad , \quad \underbrace{T_n = \frac{2\pi}{\omega_n}}_{\text{natural period}}$$

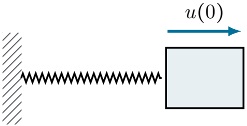
Damping makes free vibration decay

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

(Damping slightly lowers the observed free-vibration frequency)

Free vibration reveals natural structural properties

Displace, release, and observe



then remove the force

$$m\ddot{u} + ku = 0$$

$$\underbrace{\omega_n = \sqrt{\frac{k}{m}}}_{\text{natural circular frequency}} \quad , \quad \underbrace{T_n = \frac{2\pi}{\omega_n}}_{\text{natural period}}$$

How fast a structure naturally oscillates is determined by its **mass** and the **stiffness** that generates the restoring force.

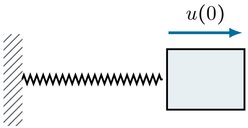
Damping makes free vibration decay

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

(Damping slightly lowers the observed free-vibration frequency)

Free vibration reveals natural structural properties

Displace, release, and observe



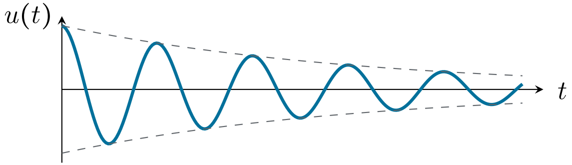
then remove the force

$$m\ddot{u} + ku = 0$$

$$\underbrace{\omega_n = \sqrt{\frac{k}{m}}}_{\text{natural circular frequency}} \quad , \quad \underbrace{T_n = \frac{2\pi}{\omega_n}}_{\text{natural period}}$$

How fast a structure naturally oscillates is determined by its **mass** and the **stiffness** that generates the restoring force.

Damping makes free vibration decay

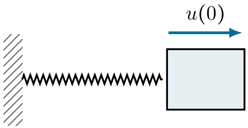


$$\omega_d = \omega_n\sqrt{1 - \zeta^2}$$

(Damping slightly lowers the observed free-vibration frequency)

Free vibration reveals natural structural properties

Displace, release, and observe



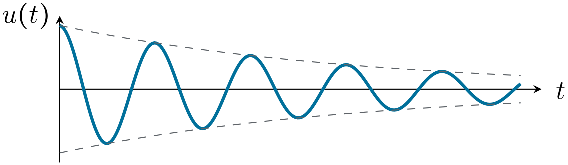
then remove the force

$$m\ddot{u} + ku = 0$$

$$\underbrace{\omega_n = \sqrt{\frac{k}{m}}}_{\text{natural circular frequency}}, \quad \underbrace{T_n = \frac{2\pi}{\omega_n}}_{\text{natural period}}$$

How fast a structure naturally oscillates is determined by its **mass** and the **stiffness** that generates the restoring force.

Damping makes free vibration decay



$$\omega_d = \omega_n\sqrt{1 - \zeta^2}$$

(Damping slightly lowers the observed free-vibration frequency)

Damping dissipates energy, so free vibration decays. Natural frequencies and damping properties of the **structure**, not of the load.

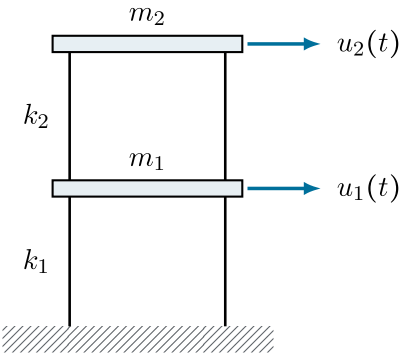
Resonance video clip

Click the image to open the video.

Contents

From one degree of freedom to many

Two-story shear-building idealization



Equations of motion

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

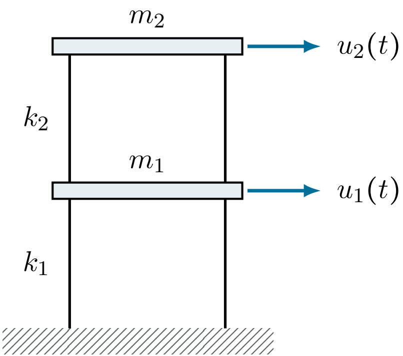
$$m_1\ddot{u}_1 + (k_1 + k_2)u_1 - k_2u_2 = 0,$$

$$m_2\ddot{u}_2 + k_2u_2 - k_2u_1 = 0.$$

Each equation depends on **both** floor displacements: the motions are **coupled**.

From one degree of freedom to many

Two-story shear-building idealization



Equations of motion

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

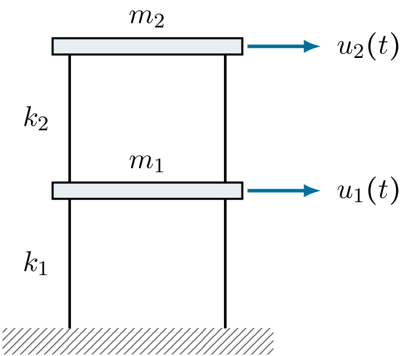
$$m_1\ddot{u}_1 + (k_1 + k_2)u_1 - k_2u_2 = 0,$$

$$m_2\ddot{u}_2 + k_2u_2 - k_2u_1 = 0.$$

Each equation depends on **both** floor displacements: the motions are **coupled**.

From one degree of freedom to many

Two-story shear-building idealization



Equations of motion

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$m_1\ddot{u}_1 + (k_1 + k_2)u_1 - k_2u_2 = 0,$$

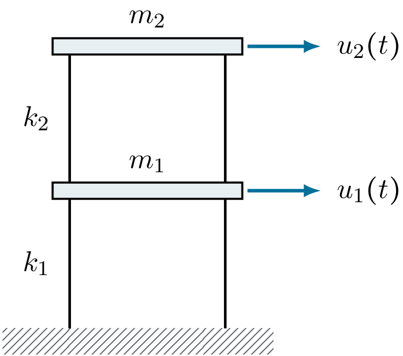
$$m_2\ddot{u}_2 + k_2u_2 - k_2u_1 = 0.$$

Each equation depends on **both** floor displacements: the motions are **coupled**.

UC Berkeley

From one degree of freedom to many

Two-story shear-building idealization



Equations of motion

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

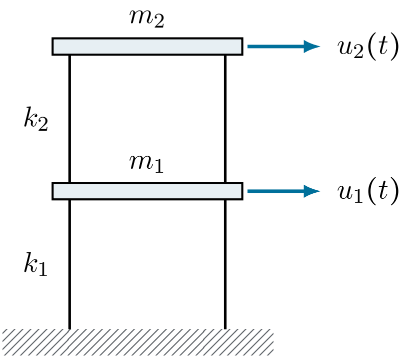
$$m_1\ddot{u}_1 + (k_1 + k_2)u_1 - k_2u_2 = 0,$$

$$m_2\ddot{u}_2 + k_2u_2 - k_2u_1 = 0.$$

Each equation depends on **both** floor displacements: the motions are **coupled**.

From one degree of freedom to many

Two-story shear-building idealization



Equations of motion

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$m_1\ddot{u}_1 + (k_1 + k_2)u_1 - k_2u_2 = 0,$$

$$m_2\ddot{u}_2 + k_2u_2 - k_2u_1 = 0.$$

Each equation depends on **both** floor displacements: the motions are **coupled**.

We seek a new coordinate system in which these equations **decouple**.

Eigenmodes decouple the equations of motion

Start from free vibration: $\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$

For free vibration, seek harmonic motion with a fixed shape:

$$\boxed{\mathbf{u}(t) = \boldsymbol{\phi} e^{i\omega t}}$$ ODE

Substitution gives a generalized eigenvalue problem:

$$\left(\mathbf{K} - \omega^2\mathbf{M}\right)\boldsymbol{\phi} = \mathbf{0} \quad \Longleftrightarrow \quad \boxed{\mathbf{K}\boldsymbol{\phi}_r = \omega_r^2\mathbf{M}\boldsymbol{\phi}_r}$$ Linear algebra

$\omega_r^2 \implies$ eigenvalue; ω_r: **natural frequency of mode** r

$\boldsymbol{\phi}_r \implies$ eigenvector: **mode shape of mode** r

The response can be expressed as a **superposition of independent modal motions**:

$$\boxed{\mathbf{u}(t) = \sum_{r=1}^{m} \boldsymbol{\phi}_r q_r(t)}$$

Each $q_r(t)$ behaves as an independent SDOF oscillator.

Eigenmodes decouple the equations of motion

Start from free vibration: $\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$

For free vibration, seek harmonic motion with a fixed shape:

$$\boxed{\mathbf{u}(t) = \phi e^{i\omega t}}$$ ODE

Substitution gives a generalized eigenvalue problem:

$$\left(\mathbf{K} - \omega^2\mathbf{M}\right)\phi = \mathbf{0} \iff \boxed{\mathbf{K}\phi_r = \omega_r^2\mathbf{M}\phi_r}$$ Linear algebra

$\omega_r^2 \implies$ eigenvalue; ω_r: **natural frequency of mode** r

$\phi_r \implies$ eigenvector: **mode shape of mode** r

The response can be expressed as a **superposition of independent modal motions**:

$$\boxed{\mathbf{u}(t) = \sum_{r=1}^{m} \phi_r q_r(t)}$$

Each $q_r(t)$ behaves as an independent SDOF oscillator.

Eigenmodes decouple the equations of motion

Start from free vibration: $\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$

For free vibration, seek harmonic motion with a fixed shape:

$$\boxed{\mathbf{u}(t) = \boldsymbol{\phi} e^{i\omega t}} \quad \textbf{ODE}$$

Substitution gives a generalized eigenvalue problem:

$$\left(\mathbf{K} - \omega^2\mathbf{M}\right)\boldsymbol{\phi} = \mathbf{0} \quad \Longleftrightarrow \quad \boxed{\mathbf{K}\boldsymbol{\phi}_r = \omega_r^2\mathbf{M}\boldsymbol{\phi}_r} \quad \textbf{Linear algebra}$$

$\omega_r^2 \Longrightarrow$ eigenvalue; ω_r: **natural frequency of mode** r

$\phi_r \Longrightarrow$ eigenvector: **mode shape of mode** r

The response can be expressed as a **superposition of independent modal motions**:

$$\boxed{\mathbf{u}(t) = \sum_{r=1}^{m} \boldsymbol{\phi}_r q_r(t)}$$

Each $q_r(t)$ behaves as an independent SDOF oscillator.

Eigenmodes decouple the equations of motion

Start from free vibration: $\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$

For free vibration, seek harmonic motion with a fixed shape:

$$\boxed{\mathbf{u}(t) = \phi e^{i\omega t}}$$ ODE

Substitution gives a generalized eigenvalue problem:

$$\left(\mathbf{K} - \omega^2\mathbf{M}\right)\phi = \mathbf{0} \qquad \Longleftrightarrow \qquad \boxed{\mathbf{K}\phi_r = \omega_r^2\mathbf{M}\phi_r}$$ Linear algebra

$\omega_r^2 \implies$ eigenvalue; ω_r: **natural frequency of mode r**

$\phi_r \implies$ eigenvector: **mode shape of mode r**

The response can be expressed as a **superposition of independent modal motions**:

$$\boxed{\mathbf{u}(t) = \sum_{r=1}^{m} \phi_r q_r(t)}$$

Each $q_r(t)$ behaves as an independent SDOF oscillator.

Eigenmodes decouple the equations of motion

Start from free vibration: $\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0}$

For free vibration, seek harmonic motion with a fixed shape:

$$\boxed{\mathbf{u}(t) = \boldsymbol{\phi} e^{i\omega t}}$$ ODE

Substitution gives a generalized eigenvalue problem:

$$\left(\mathbf{K} - \omega^2 \mathbf{M}\right) \boldsymbol{\phi} = \mathbf{0} \quad \Longleftrightarrow \quad \boxed{\mathbf{K}\boldsymbol{\phi}_r = \omega_r^2 \mathbf{M}\boldsymbol{\phi}_r}$$ Linear algebra

$\omega_r^2 \implies$ eigenvalue; ω_r: **natural frequency of mode** r

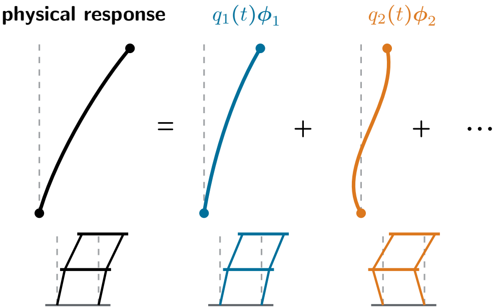
$\boldsymbol{\phi}_r \implies$ eigenvector: **mode shape of mode** r

The response can be expressed as a **superposition of independent modal motions**:

$$\boxed{\mathbf{u}(t) = \sum_{r=1}^{m} \boldsymbol{\phi}_r q_r(t)}$$

Each $q_r(t)$ behaves as an independent SDOF oscillator.

The structural response is a superposition of modes

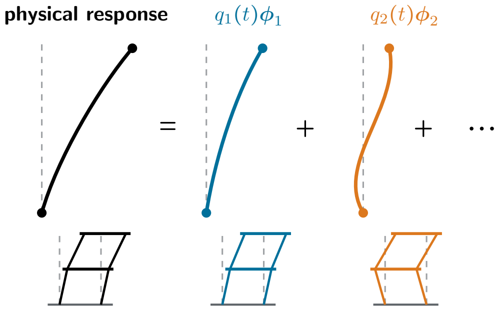


$$\boxed{\mathbf{u}(t) = \mathbf{\Phi}\mathbf{q}(t) = q_1(t)\phi_1 + q_2(t)\phi_2}$$

Interpretation

- ϕ_r: a fixed spatial deformation pattern,
- $q_r(t)$: how strongly that pattern is present at time t,
- each $q_r(t)$ follows an SDOF-like equation.

The structural response is a superposition of modes

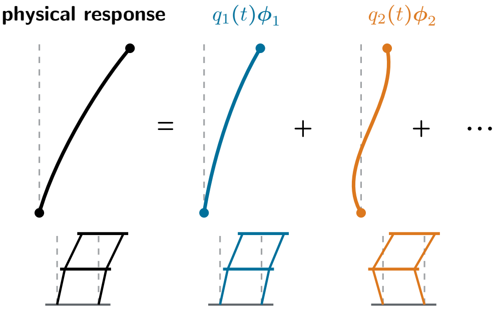


$$\boxed{\mathbf{u}(t) = \mathbf{\Phi}\mathbf{q}(t) = q_1(t)\phi_1 + q_2(t)\phi_2}$$

Interpretation

- ϕ_r: a fixed spatial deformation pattern,
- $q_r(t)$: how strongly that pattern is present at time t,
- each $q_r(t)$ follows an SDOF-like equation.

The structural response is a superposition of modes



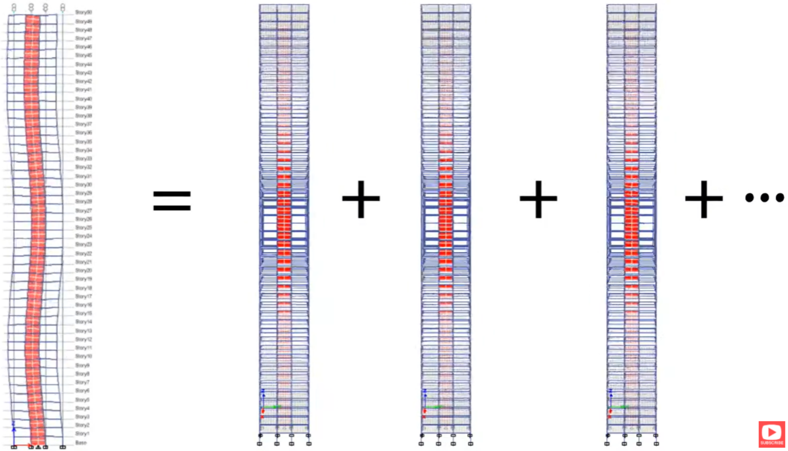
$$\boxed{\mathbf{u}(t) = \mathbf{\Phi}\mathbf{q}(t) = q_1(t)\phi_1 + q_2(t)\phi_2}$$

Interpretation

- ϕ_r: a fixed spatial deformation pattern,
- $q_r(t)$: how strongly that pattern is present at time t,
- each $q_r(t)$ follows an SDOF-like equation.

Modal coordinates turn a coupled linear system into simpler modal oscillators.

Mode superposition video clip



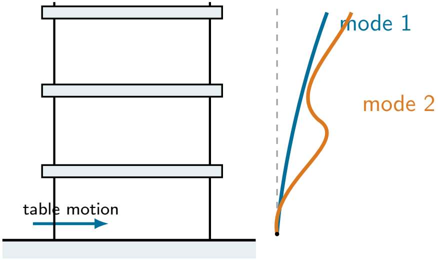
Click the image to open the video.

Contents

Shake-table demonstration

What do the mode shapes actually look like?

Let's find out on the shake table.



Contents

Concepts to review before CE 225

Differential equations

- second-order linear ODEs (covered in the ODE primer),
- homogeneous and particular solutions; initial conditions,
- sinusoidal and exponential trial solutions.

Linear algebra

- matrix multiplication and linear systems,
- eigenvalues and eigenvectors.

Structural analysis

- selecting degrees of freedom for a structural model,
- element stiffness matrices and assembly.

Three takeaways

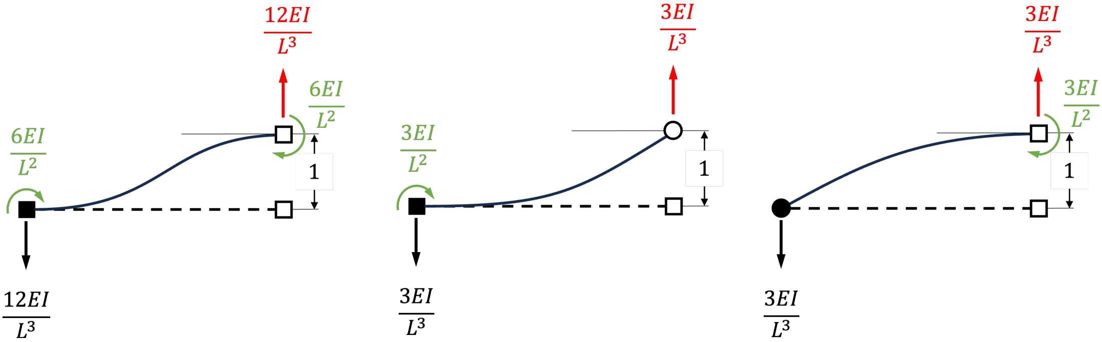
1. Structural dynamics is structural analysis with **inertia, damping, and time**.
2. **Mass and stiffness** set the natural frequency; **damping** makes vibration decay.
3. A linear structure can be decomposed into **eigenmodes**, each behaving like an SDOF oscillator.

CE 225 assumes frame-element stiffness

Most of CE 225 is self-contained, but one result from structural analysis is often used without re-derivation:

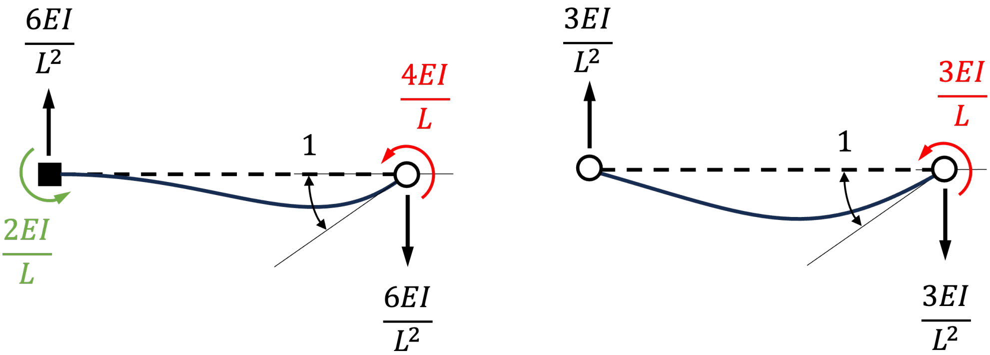
the frame-element stiffness coefficients

Reference: unit translations



How to read: (For the left one) To move the right node upward by a unit displacement while keeping the rotations at both nodes fixed, the beam requires a shear force of magnitude $\dfrac{12EI}{L^3}$ and end moments of magnitude $\dfrac{6EI}{L^2}$.

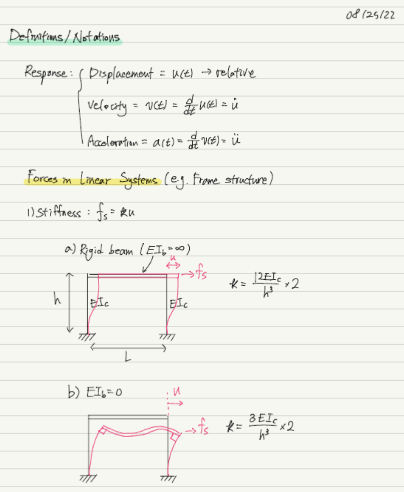
Reference: unit rotations



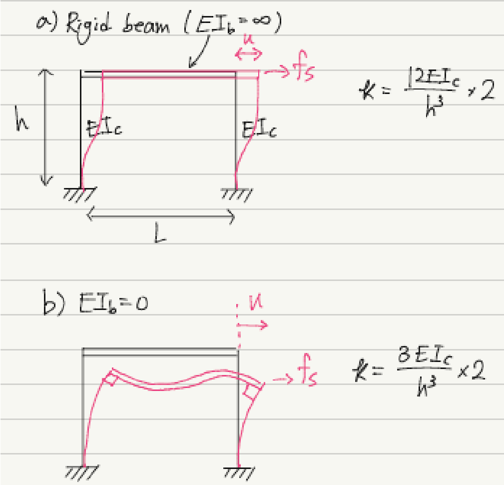
Take a photo of these two slides or copy the coefficients into your notes. You will use them later in CE 225 when assembling $\mathbf{K}$.

A real example from CE 225

My notes from the first CE 225 lecture



Stiffness in two limiting cases



You are ready for CE 225.

Appendix: when fixed linear modes stop being exact

Linear structure

$$\mathbf{f}_{\text{int}} = \mathbf{K}\mathbf{u}$$

fixed $\mathbf{K}$
fixed mode shapes
exact modal superposition

Nonlinear structure

$$\mathbf{f}_{\text{int}} = \mathbf{f}_{\text{int}}(\mathbf{u}, \text{history})$$

tangent stiffness changes
fixed modes are approximate
direct integration may be needed

In this primer, the modal viewpoint refers primarily to **linear structural dynamics**.